Chronosequence-Anchored Gradient-Boosted Ensembles with Anchor Shrinkage for Global Land Ecosystem Forecasting under SSP Scenarios

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***Abstract*—The IEEE Big Data Cup 2026 CarbonGlobe challenge asks for the year-40 height and aboveground biomass (AGB) of 6,171 held-out sites, at 15 forest seed ages, under the SSP126 and SSP585 climate scenarios. Only the end state is scored, so a model need not reproduce the full yearly trajectory. We therefore avoid autoregressive rollout and predict the year-40 state directly. Our pipeline (i) builds a *chronosequence anchor* by reading each site's own initial-state curve at seed age $a+40$; (ii) engineers 1,159 features from eight climate summaries of 136 drivers, ecological ratios, site carrying-capacity statistics and age-dynamics terms; (iii) trains four gradient-boosting and random-forest engines on the anchor residual, the direct target and a log-ratio target, and combines them with convex weights fitted by SLSQP; and (iv) shrinks the ensemble toward the anchor with a scenario-dependent coefficient and clips it to a physically motivated band around the anchor. The final model is refitted on all labelled sites and produces 185,130 valid predictions. The best recorded public leaderboard score (mean squared relative error, lower is better) is 0.095. We also discuss the limitations of our validation protocol and several design choices that remain untested.**

***Index Terms*—CarbonGlobe, ecosystem emulation, chronosequence, space-for-time substitution, gradient boosting, ensemble learning, SSP scenarios, distribution shift**

I. Introduction

Process-based ecosystem models such as the Ecosystem Demography (ED) model simulate forest growth, mortality and carbon exchange from climate and soil forcing [3], but running them globally for many climate scenarios is computationally expensive. Machine-learning emulators offer a cheap substitute. CarbonGlobe is an ML-ready benchmark for this purpose, with 40 years of monthly environmental forcing, 136 input variables, 15 forest-age conditions and 54,152 land sites at 0.5° resolution, together with ED outputs for seven ecosystem variables [1]. The IEEE Big Data Cup 2026 challenge [2] requests forecasts at 6,171 held-out sites under the low-emission SSP126 and high-emission SSP585 scenarios [4].

Two properties of the task shape our approach. First, only the year-40 height and AGB are scored, so intermediate years are irrelevant to the metric. Second, the test climates are future climates and may lie outside the range of the present-day training climate. A model that leans heavily on climate input can therefore extrapolate poorly.

Each site is simulated from 15 seed ages (1 to 500 years). For old stands the state changes slowly, so the site's own initial-state values at older seed ages already indicate what a stand of age $a + 40$ looks like at that location. This is the chronosequence, or space-for-time, argument [5] applied along the age axis of a single site. We use it as an anchor and let boosted-tree models learn how the true year-40 state departs from it.

A. Contributions

- A non-autoregressive formulation that predicts the year-40 state directly and uses the site's own initial-state curve at age $a + 40$ as a climate-free anchor.
- A 1,159-dimensional feature set combining climate summaries, ecological ratios, site carrying-capacity ("headroom") statistics, chronosequence features and age-dynamics terms.
- A convex multi-engine ensemble (LightGBM residual, direct and log-ratio models, and a random-forest engine) whose weights are fitted on a held-out validation split.
- A scenario-dependent shrinkage toward the anchor and a bounded-clipping rule that keep predictions inside a physically plausible band.

II. Related Work

A. Emulating process-based ecosystem models

The CarbonGlobe benchmark provides recurrent and attention-based baselines that forecast multi-decadal trajectories from forcing and initial states [1]. Such sequence emulators are usually trained and validated on random held-out sites. Our method differs in that it targets only the scored year-40 state and uses the age axis as a source of information about the target itself.

B. Space-for-time substitution

Ecologists infer temporal change from stands of different ages observed at a single time [5]. We apply the same idea inside one simulated site: the 15-age initial-state curve acts as a chronosequence from which the state at a later age can be interpolated.

C. Gradient boosting and random forests

Gradient-boosted trees [6], [7] are strong on tabular data and do not extrapolate beyond the range of the training targets, which is a useful property under distribution shift. Random

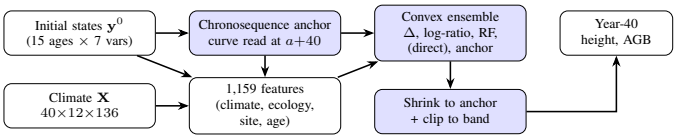


Fig. 1. Proposed pipeline. Shaded boxes carry the forecast.

forests [8] average many decorrelated trees and give smoother predictions. We use both families, and we combine them with convex weights.

D. Growth-curve priors

Saturating growth functions such as the Richards (Chapman–Richards) family [9] motivate our "headroom" and asymptotic-potential features: a stand approaches a site-specific maximum, and the remaining distance to that maximum bounds the achievable growth.

III. Problem and Data

A. Task and metric

For site s and seed age a_j ($j = 1, \ldots, 15$; ages $1, 10, 20, 30, 50, 70, 100, 150, 200, 250, 300, 350, 400, 450, 500$), the input is an initial state $\mathbf{y}^0_{s,j} \in \mathbb{R}^7$ (height, AGB, soil carbon, LAI, GPP, NPP, Rh) and a forcing tensor $\mathbf{X}_s \in \mathbb{R}^{40\times12\times136}$. The output is the year-40 height $h_{s,j}$ and AGB $b_{s,j}$, divided by their global means $\mu_h = 10.90645$ and $\mu_b = 3.39567$. The score is

$$S = \tfrac{1}{2}\left(\mathrm{MSE}_h + \mathrm{MSE}_b\right), \tag{1}$$

the mean squared relative error. The test set has 6,171 sites, 15 ages and two scenarios, giving 185,130 rows.

B. Training data

The global training data hold, for each of the 54,152 sites and each seed age, a trajectory of years 0 to 40. We take $\mathbf{y}^0$ from year 0 and the target from year 40, and pair them with the site's climate tensor. This gives $54{,}152 \times 15 = 812{,}280$ site–age rows when all labelled sites are used.

IV. Method

A. Overview

Fig. 1 shows the pipeline. For every test site and seed age we compute a chronosequence anchor, build the 1,159-dimensional feature vector, obtain predictions from four engines, blend them, shrink the blend toward the anchor and clip the result.

B. Chronosequence anchor

For variable $k \in \{\text{height}, \text{AGB}\}$, let $y^0_{s,a,k}$ be the initial value of site s at seed age a. The anchor for seed age a_j is the linear interpolation of the 15-point curve at age $a_j + 40$, with the query clipped to $[1, 500]$:

$$c_{s,j,k} = \mathrm{interp}_a\big(y^0_{s,a,k};\ \min(a_j + 40,\ 500)\big). \tag{2}$$

The anchor uses no climate information and is available at test time because the whole age curve of a site is provided. It supplies the part of the year-40 state that the initial-state curve already contains. The same interpolation is also computed for NPP and LAI and used as features.

C. Feature engineering

Each row has 1,159 features in five groups.

1) **Climate (1,088).** For each of the 136 drivers, annual means are standardised with the global statistics and summarised by eight quantities: the 40-year mean, the mean of the first five years, the mean of the last five years, their difference, the standard deviation of annual means, the mean within-year (seasonal) standard deviation, the linear trend, and a mean weighted toward later years.
2) **Initial state (16).** The seven raw values, their $\log(1+\cdot)$ transforms, and height and AGB scaled by μ_h, μ_b.
3) **Ecological ratios (11).** Carbon-use efficiency (NPP/GPP), net ecosystem productivity (NPP$-$Rh) and its ratio to AGB, biomass density, an effective specific leaf area (GPP/LAI), soil turnover (Rh/soil), AGB/soil, relative growth rate, an allometric height–biomass term, total carbon and AGB$-$height.
4) **Site carrying-capacity and chronosequence (20).** The site-wide maxima and means over the 15 ages (height, AGB, NPP, LAI), headroom (site maximum minus current value), saturation ratios, the anchors $c_{s,j,k}$ for height, AGB, NPP and LAI, and the differences between anchor and current value.
5) **Age dynamics (24).** Normalised age and its transforms, the future-age ratio $(a + 40)/a$ and its logarithm, exponential decay terms at several time scales, per-year rates, asymptotic-potential terms (headroom multiplied by $1 - \exp\big(-40/(a + 10)\big)$), anchor-to-current ratios and age–anchor interactions.

D. Targets and engines

For each target we train four regressors on the scaled quantities. With g_k the global mean of target k and $z_{s,j,k}$ the year-40 value:

$$\text{Residual } (\Delta): \quad (z - c)/g_k, \tag{3}$$

$$\text{Direct:} \quad z/g_k, \tag{4}$$

$$\text{Log-ratio:} \quad \log(1 + z) - \log(1 + c), \tag{5}$$

$$\text{Random forest:} \quad (z - c)/g_k. \tag{6}$$

The first three use LightGBM gradient boosting [7] with 191 leaves, learning rate 0.025, column sampling 0.72–0.78, ℓ_1/ℓ_2 regularisation and early stopping on a validation split (patience 220, up to 4,500 trees). The random-forest engine uses LightGBM in random-forest mode (127 leaves, depth 16, 1,000 trees). All predictions are converted back to the scaled target space; the log-ratio output is first clipped to $[-3, 2.5]$.

E. Convex ensemble

Let $\mathbf{p} \in \mathbb{R}^5$ collect the predictions of the Δ, direct, log-ratio and random-forest engines and the anchor itself. Weights $\mathbf{w}$

are found per target by minimising the validation mean squared error under $w_i \geq 0$ and $\sum_i w_i = 1$ (SLSQP):

$$\mathbf{w}^\star = \arg\min_{\mathbf{w}\in\Delta^4} \ \tfrac{1}{n}\sum_i \left(\mathbf{w}^\top \mathbf{p}_i - \tilde{z}_i\right)^2. \tag{7}$$

At inference the weight of the direct engine is set to zero and the remaining weights are renormalised, so that the final forecast is built around the anchor rather than around a climate-only prediction.

F. Anchor shrinkage and bounding

The blended prediction $\hat{z}^{\text{bl}}$ is shrunk toward the scaled anchor $\hat{c} = \max(c, 0)/g_k$ with a scenario-dependent coefficient α:

$$\hat{z}^{\text{sh}} = (1-\alpha)\,\hat{z}^{\text{bl}} + \alpha\,\hat{c}, \qquad \alpha = \begin{cases} 0.40 & \text{SSP126} \\ 0.85 & \text{SSP585.} \end{cases} \tag{8}$$

The larger coefficient for SSP585 reflects the larger expected climate shift under the high-emission scenario. The values are set by hand and are not fitted. Predictions are finally clipped to a band around the anchor,

$$\hat{z} = \text{clip}\Big(\hat{z}^{\text{sh}},\ 0.8\,\hat{c},\ \max\big(1.05\,m_k/g_k,\ 1.25\,\hat{c}\big)\Big), \tag{9}$$

where m_k is the maximum value of variable k over the 15 ages of the site. The band prevents implausible growth or loss relative to the site's own chronosequence.

G. Training protocol

Models are first trained on the training sites, with the validation sites used for early stopping and for fitting the ensemble weights. The final models are then refitted on all labelled sites (training, validation and the additional labelled split), using a memory-limited voting ensemble of LightGBM, XGBoost [6] and histogram gradient boosting (300 trees, depth 8) for each of the first three engines, and LightGBM plus XGBoost random forest for the fourth. During training only mild augmentation is applied: Gaussian noise (standard deviation 0.005) on the climate input and a multiplicative perturbation in $[0.995, 1.005]$ on the initial state. The target is left unchanged.

V. Experimental Setup

A. Data and software

We use the CarbonGlobe global arrays (54,152 sites) and the competition test files for SSP126 and SSP585. The pipeline is implemented in Python with NumPy, pandas, LightGBM, XGBoost and scikit-learn, and was run on Kaggle. Memory use is controlled by memory-mapping the arrays and building features in batches of 2,000 sites.

B. Evaluation

The primary metric is the mean squared relative error in (1), reported by the competition's public leaderboard. During development we also report a validation score on held-out labelled sites (the same metric computed on the validation split).

TABLE I
Model performance and public leaderboard results (lower is better).

Evaluation	Height	AGB
Validation MSE (current code)	0.022335	0.047294
Overall validation scaled-MSE	0.034814	
Earlier HGB public score	0.238	
Current best public score	**0.095**	
Current latest displayed score	0.101	

TABLE II
Public leaderboard error reduction from the earlier baseline to the current approach (lower is better).

Metric	**Best current**	**Latest submission**
Earlier HGB score (reference)	0.238	0.238
Current public score	**0.095**	0.101
Absolute score reduction	**0.143**	0.137
Relative score reduction	**60.1%**	57.6%

VI. Results

A. Leaderboard and validation performance

Table I summarizes the observed competition scores and the validation performance logged by the current notebook. The earlier histogram gradient boosting submission scored 0.238 on the public leaderboard. The best recorded score for the current approach is 0.095, while the latest submission shown in the supplied leaderboard screenshot scored 0.101. Since the competition metric is lower-is-better, the best score represents an improvement of approximately 60.1% relative to the earlier 0.238 score. The 0.034814 validation value is the notebook's overall scaled-MSE diagnostic and should not be interpreted as the public leaderboard score.

B. Error reduction: before versus after

To make the change in error explicit, Table II compares the earlier Histogram Gradient Boosting (HGB) submission with the current ensemble using the public leaderboard metric. The competition score is lower-is-better; absolute reduction is computed as $S_{\text{before}} - S_{\text{after}}$, and relative reduction divides this difference by the earlier score. The earlier per-target validation errors were not present in the supplied notebook output, so they are not estimated here. The current validation MSE values are reported separately in Table I and must not be compared directly with leaderboard scores.

C. Submission output and prediction statistics

The attached `submission.csv` contains 185,130 rows and the columns `id`, `height`, and `agb`. All height and AGB values are non-negative and there are no missing predictions. Table IV reports descriptive statistics calculated directly from the supplied CSV; values are in the normalized/scaled target units used by the submission.

Table III directly contrasts the earlier histogram-gradient-boosting baseline with the current chronosequence-anchored approach. The current final refit uses a voting ensemble of

TABLE III
BEFORE-AND-AFTER COMPARISON OF THE MODELLING APPROACHES.

Aspect	Earlier approach	Current approach
Models	Histogram Gradient Boosting	LightGBM + XGBoost + HistGradientBoosting voting ensembles; LightGBM + XGBoost for RF engine
Features	Earlier baseline feature set	1,159 climate, ecological, site, and age features
Post-processing	Baseline prediction	Chronosequence anchor, scenario-dependent shrinkage, and clipping
Best public score	0.238	**0.095**
Latest displayed score	–	0.101
Absolute reduction	–	0.143 vs. baseline
Relative reduction	–	60.1% vs. baseline

TABLE IV
SUMMARY STATISTICS OF PREDICTIONS IN THE SUBMITTED CSV (SCALED UNITS).

Statistic	Height	AGB
Count	185130	185130
Mean	0.9040	0.9639
Standard deviation	0.8559	1.4047
Minimum	0.0000	0.0000
25th percentile	0.0715	0.0000
Median	0.6235	0.1862
75th percentile	1.7349	1.5121
Maximum	2.7296	6.7653

LightGBM, XGBoost, and scikit-learn HistGradientBoosting for the residual, direct, and log-ratio engines; the random-forest engine uses LightGBM and XGBoost. The current result is the best recorded public leaderboard score. Since only the earlier score and current leaderboard scores are available, this comparison reports competition scores rather than claiming a controlled component-wise ablation.

The prediction distributions are similar across SSP126 and SSP585 at the aggregate level: mean height is approximately 0.904 in both scenarios, and mean AGB is approximately 0.9639 for SSP126 and 0.9649 for SSP585. This similarity is consistent with the strong anchor shrinkage used by the current pipeline, particularly for SSP585. It does not by itself establish equal scenario-level accuracy because test labels are unavailable.

VII. DISCUSSION

A. Why an anchor

Because the anchor needs only the site's own initial-state curve, it is unaffected by the climate shift between the training and test data. The learned engines therefore only need to model a bounded correction, and the clipping band limits the damage when they extrapolate badly.

TABLE V
EXAMPLE ROWS FROM THE GENERATED SUBMISSION.

Submission ID	Height	AGB
ssp126_0_1	0.007755	0.000000
ssp126_0_10	0.002394	0.000000
ssp126_0_20	0.000000	0.000000
ssp126_0_30	0.000000	0.000000
ssp126_0_50	0.000000	0.000000
ssp126_0_500	0.000000	0.000000
ssp585_0_1	0.007683	0.000000
ssp585_0_10	0.002322	0.000000
ssp126_666_300	0.500960	1.144023
ssp585_495_300	0.318355	0.317977
ssp126_6000_1	0.703914	0.063543
ssp585_6170_500	0.493880	1.337991

B. Role of shrinkage

Tree ensembles trained on present-day climate saturate rather than extrapolate, so their disagreement with the anchor on shifted inputs is a source of risk. Shrinking toward the anchor trades variance for bias. Our larger coefficient for SSP585 encodes the prior that this scenario is further from the training climate; we did not tune it against labels, and a systematic study is left for future work.

VIII. THREATS TO VALIDITY

Validation optimism. The validation split is used both for early stopping and for fitting the ensemble weights, so the reported validation score is optimistic.

Refit mismatch. The ensemble weights are fitted using the early-stopped models (up to 4,500 trees), but the final predictions use memory-limited refits (300 trees, depth 8, voting ensembles). The weights are therefore not exactly optimal for the deployed models.

Hand-set post-processing. The shrinkage coefficients (0.40, 0.85), the band (0.8, 1.25, 1.05) and the removal of the direct engine at inference were chosen by judgement and are not validated against held-out labels. We have no ablation isolating their effects.

No distribution alignment. We do not align the test initial states or climate summaries to the training distribution, so differences between test and training inputs are only handled implicitly through the anchor, shrinkage and clipping.

Single run. Each score comes from one run, so small differences cannot be distinguished from seed variation.

Simulator target. The labels are ED outputs; agreement with ED is not agreement with field observations.

IX. CONCLUSION AND FUTURE WORK

We presented a non-autoregressive pipeline for the CarbonGlobe year-40 forecasting task. A chronosequence anchor read from each site's own initial-state curve is combined with a feature-rich, multi-engine boosted ensemble, then shrunk toward the anchor and clipped to a plausible band. The submission is complete and valid (185,130 rows), with a best recorded public leaderboard score of **0.095**(the latest screenshot shows 0.101). Future work includes (i) ablations of the anchor, each engine, the shrinkage coefficients and the clipping band; (ii)

fitting the shrinkage coefficient on shift-aware validation folds rather than setting it by hand; (iii) aligning test inputs to the training distribution; and (iv) a validation protocol that tracks the leaderboard under climate shift.

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